

ASSIGNMENT - 3

Problems on Fourier series:

Obtain the Fourier Series for the following:

1. $f(x) = e^x$ in $(-l, l)$

Ans: $f(x) = \frac{\sinh l}{l} - \frac{2\sinh l}{l} \sum_{n=1}^{\infty} \frac{(-1)^n}{1^2 + n^2} (-\cos nx + n \sin nx)$

2. $f(x) = \begin{cases} 0 & \text{in } (-\pi, 0) \\ \sin x & \text{in } (0, \pi) \end{cases}$ Deduce $\frac{\pi-2}{4} = \frac{1}{1.3} - \frac{1}{1.5} + \frac{1}{5.7} \dots$

Ans: $a_0 = -\frac{2}{\pi}$ $a_n = \frac{-1}{\pi(n^2-1)} \{1 - (-1)^n\}$ where $n \neq 1$, $b_n = 0$

3. $f(x) = x \cos x$ in $(-\pi, \pi)$

Ans: $a_0 = 0$ $a_n = 0$, $b_n = \frac{2n(-1)^n}{n^2-1} (n \neq 1)$